VULCAN OMNIPRO 220 SUPPORT AGENT

Architecture, Design Decisions, and Evaluation

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1. The Product

The Vulcan OmniPro 220 is an industrial multiprocess welder. Multiprocess means it's 4 welders in 1 machine. Instead of buying separate dedicated machines, this single unit switches between all 4 major arc welding processes: MIG, Flux-Core, TIG, and Stick.

On a single-process machine you'd be locked into one of those.

We built a technical support agent that answers questions about this machine. The target user: someone standing in their garage, phone propped on a workbench, trying to get their welder working. They need fast, correct answers grounded in the owner's manual.

2. How the Agent Works

The agent is a RAG system built on the Claude Agent SDK. It uses 7 custom tools registered through an MCP server, backed by structured JSON, BM25-indexed markdown, and 300 DPI page images from a 48-page manual.

Request Flow

Step	What Happens
1. User sends message	Frontend sends message + process context + voltage + session_id via POST /api/chat
2. Context injection	Server prepends [User has selected process: MIG, voltage: 240V] to the message
3. Session lookup	Server finds or creates a persistent ClaudeSDKClient for this session_id
4. Agent reasoning	Claude Haiku reads the system prompt, decides which tools to call
5. Tool execution	Agent calls 1-5 tools (lookup_specs, search_manual_text, etc.)
6. Response generation	Agent synthesizes tool results into a markdown response with citations
7. SSE streaming	Response streams to the frontend via Server-Sent Events

Three architectural decisions define the system:

Structured data for safety-critical values. Specs and troubleshooting data live in curated JSON files, not extracted at query time. The agent calls lookup_specs to read exact numbers from a deterministic source. This eliminates hallucination risk for duty cycles, amperage ranges, and polarity configurations.

BM25 for text search, not vector embeddings. The manual is 1,648 lines. Too small for a vector database to add value. BM25 (rank_bm25 library) chunks the markdown by section headers (111 chunks) and returns relevance-scored results. Fast, simple, no external dependencies.

Page images as a separate modality. 22 critical pages have hand-written text descriptions. The agent can also serve the actual PNG images via URL. This handles both "explain what's on page 13" and "show me page 13."

2.1 Knowledge Base

Built from 3 source documents processed through a two-stage pipeline: PDF rasterization at 300 DPI (pdf2image), then markdown extraction (IBM Docling, 97.9% table accuracy).

File	Size	Purpose
`specs.json`	6.7 KB	Structured specs for all 4 processes at 120V/240V. Duty cycles, amperage ranges, wire sizes, polarity configs. Single source of truth for numbers.
`troubleshooting.json`	14.6 KB	Problem/Cause/Solution matrices. Three sections: mig_flux_cored, tig_stick, weld_diagnosis. Structured for direct lookup.
`page_index.json`	19.2 KB	Metadata for all 51 pages: sections, topics, content types, critical_data flags. Powers the search_pages_by_topic tool.
`owner-manual.md`	1,648 lines	Full text extracted by Docling. Chunked into 111 sections by ## headers for BM25 indexing.
`pages/*.png`	51 files	300 DPI rasterizations of every page. Served as static files via /pages/ route.
`page_descriptions/*.md`	22 files	Hand-written descriptions of critical visual pages: specs tables, polarity diagrams, LCD screens, weld diagnosis photos, wiring schematic.

specs.json contains exact manufacturer values. The agent reads these through lookup_specs rather than extracting them from markdown at query time. Hallucination of numerical specs is structurally impossible: the numbers come from a curated JSON file, not from LLM interpretation of OCR'd text.

2.2 Tool Design

7 tools, each built for a specific query pattern. All registered as read-only and instrumented with Logfire spans.

Tool	What It Does and Why
`lookup_specs`	Returns exact specs from specs.json. Filters by process name, cutting token usage by ~75%. The most important anti-hallucination tool: the LLM never estimates specs, it reads curated JSON.
`lookup_troubleshooting`	Returns structured Problem/Cause/Solution data. Routes to the right section (mig_flux_cored, tig_stick, or weld_diagnosis) by keyword. Pre-structured data keeps the agent from inventing diagnostic logic.
`get_polarity_quick_reference`	Returns polarity config (DCEP/DCEN, which cable in which socket) per process. Polarity errors are dangerous: wrong polarity means bad welds and potential equipment damage. A dedicated tool ensures the correct config every time.
`search_manual_text`	BM25 relevance-ranked search across 111 manual sections. Returns top 5 results with scores. Upgraded from substring matching, which couldn't handle natural language queries like "wire feed tension for thin steel."
`get_page_description`	Returns a hand-written text description of a visual page. Available for 22 critical pages. Text descriptions cost fewer tokens than images, so the agent reads the description first, then decides whether to show the actual image.
`get_page_image`	Returns a URL path to the page PNG with a markdown embed instruction. Changed from base64 to URL after the frontend couldn't render base64 in the SSE stream. Static serving is faster and cacheable.
`search_pages_by_topic`	Searches page_index.json to find which pages cover a topic. A directory lookup: the agent uses this to navigate the manual structure before fetching content.

From eval runs, the average query uses 2-5 tool calls. Simple spec questions ("duty cycle at 200A") need 1 call. Complex troubleshooting ("TIG arc wanders") needs 5+ as the agent cross-references troubleshooting, polarity, and manual search.

2.3 System Prompt

~4,900 characters. Defines identity, tool usage, query routing, response format, and artifact policy. Every section was refined through testing.

Section	Size	Purpose
Identity + scope	2 sentences	What the agent does and which product it covers
Tool descriptions	7 bullets	One-line description per tool with when-to-use guidance
Query routing	6 rules	Explicit mapping from query type to tool. Cuts unnecessary calls.
Rules	9 numbered	Hard constraints: never guess values, cite sources, admit limits, keep responses short
Code generation	3 sections	When to generate artifacts (opt-in only), when not to, and how
Citation format	With examples	Exact bracket format [p.7]. Good/bad examples of grouped vs repetitive citation.

The 9 rules, annotated:

1. Never guess technical values. If lookup_specs doesn't have it, the agent says "I don't have that information." A wrong duty cycle could cause equipment damage or fire.

2. Show diagrams when they help. Polarity setups and cable routing are spatial. Text descriptions are inferior. The agent fetches page images for visual questions.

3. Safety first, bold warnings. Welding is dangerous. Safety-relevant information must be visually prominent.

4. Cross-reference when needed. Many real questions require combining specs + troubleshooting + polarity data. The agent should use multiple tools.

5. Never ask clarifying questions (unless genuinely ambiguous about which PROCESS). For everything else, search first. Added after the agent asked "What do you mean?" for "american air fitting" instead of searching for part #27.

6. Be practical and direct. The user is in their garage. Skip jargon, skip background, give the answer.

7. Keep responses SHORT. 2-4 sentences when possible. No "let me know if you need anything else." Added after observing over-explanation on simple spec questions.

8. Cite in bracket format. [p.7], [p.13], not [manual] or [specs]. One citation per section, not per row. Prevents spam while maintaining traceability.

9. Admit limits honestly. Don't fabricate answers about competitor products, aftermarket parts, or topics outside the manual. This rule drives the hallucination test suite.

Query routing cuts tool calls from ~5.5 to ~3 per query

Without explicit routing, the agent trial-and-errors its way to the right tool. We observed Haiku calling search_manual_text 4-5 times with slight keyword variations before finding the right section. The routing table in the system prompt tells it which tool to reach for first.

Query Pattern	First Tool	Example Keywords
Spec/number questions	`lookup_specs`	duty cycle, amperage, wire size, voltage
Problem/symptom	`lookup_troubleshooting`	arc unstable, wire not feeding, porosity
Cable/socket/polarity	`get_polarity_quick_reference`	which socket, ground clamp, DCEN/DCEP
"How do I..." setup	`search_manual_text`	procedures, installation, assembly
Visual/diagram	`search_pages_by_topic ->`	"show me", "draw", "diagram of"

	`get_page_image`	
"Where in the manual"	`search_pages_by_topic`	page number, section, location

2.4 Anti-Hallucination Design

For a welding manual agent, hallucination isn't just wrong. It's dangerous. A fabricated duty cycle could cause a fire. A wrong polarity could damage equipment. Five layers of defense:

1. Structured data store: Specs and troubleshooting are curated JSON. The agent reads deterministic values. The strongest guarantee.

2. System prompt guardrails: Rule 1 ("never guess") and Rule 9 ("admit limits"). These reduce but don't eliminate risk.

3. Query routing: Directing spec questions to lookup_specs prevents the agent from answering from memory or misinterpreted markdown.

4. Citation requirements: Forcing [p.7] citations makes the agent trace claims to sources. If it can't cite, it's more likely to say "I don't know."

5. Eval-driven verification: 8 adversarial test cases (EVAL-H01 through H08) specifically test for hallucination. 8/8 passed.

> Result: 8/8 adversarial hallucination tests passed. The agent correctly refused out-of-range specs (250A on 120V MIG), nonexistent features (AC TIG, wire preheater), competitor data (Lincoln Power MIG 210), and corrected a user's wrong polarity assumption (DCEN for MIG).

3. The Interface

Designed for a welder in their garage, phone or laptop on the workbench. The interface needs to be fast to scan, unambiguous, and surfacing the right context without being asked.

3.1 Three-panel layout

A chat-only interface forces the user to ask for context they could have at a glance. Welders switch between processes frequently (MIG to flux-cored for outdoor work, TIG for thin stainless). A single-panel chat would mean re-asking basic questions every time.

Panel	Contains	Why
Left (220px)	Process selector with duty cycle indicators	Welders think in terms of process first. Selecting MIG vs TIG changes which specs are relevant.
Center (flex)	Chat with markdown + artifact rendering	The core interaction. Tables, code blocks, blockquotes, and sandboxed iframes for interactive diagrams.
Right (280px)	Live context: duty cycles, polarity, specs	Updates on process/voltage switch. Eliminates the most common repeat questions. No LLM call needed.

3.2 Interaction details

Voltage toggle in the header. 120V vs 240V is a machine-level setting, not a process setting. It lives in the header (global scope) rather than in the left panel (process scope). Toggle buttons with active state highlighting make the current selection unambiguous.

Process selection with visual hierarchy. Each item shows the process name, a subtitle (e.g., "GMAW" for MIG), and a large number. The active process gets a border highlight and white background with subtle shadow.

Typing indicator. Three animated dots (0.2s stagger) signal the agent is working. Tool calls can take 10-20 seconds. Without an indicator, users assume the app is broken.

Artifact rendering in iframes. HTML code blocks from the agent render in sandboxed iframes within the chat. Interactive elements (sliders, click handlers) can't affect the parent page. Each artifact is self-contained with inline CSS and JS.

Voice input. A microphone button uses the Web Speech API. The target user may have welding gloves on or dirty hands. Typing on a phone keyboard is impractical. The button pulses red when listening.

4. How We Tested It

We started by reading real user questions on Reddit (r/Welding, r/harborfreight) before defining any metrics. Failure categories should emerge from actual usage, not imagination.

We fed discussion threads into an agent for topic modelling to find natural dimensions of variation.

4.1 Three dimensions

We defined 3 axes targeting where a welding manual RAG agent is most likely to fail. Together they create 7 x 6 x 4 = 168 possible tuples. We selected 58 meaningful combinations.

Dimension 1: Query Type

What the user needs from the agent

Value	Why It Matters
`troubleshooting`	Requires matching symptoms to structured problem/cause/solution data
`spec_lookup`	Must return exact numbers, never estimate
`setup_configuration`	Multi-step procedures, often needs visual diagrams
`process_guidance`	Advisory: must stay within manual scope, not give opinions
`comparison_purchase`	Out-of-scope trap: agent must not fabricate competitor data
`safety`	Highest stakes: incorrect safety info is dangerous
`weld_diagnosis`	Requires interpreting visual weld quality photos/descriptions

Dimension 2: Welding Process

Which welding process is relevant

Value	Why It Matters
`MIG`	Most common process, most questions asked
`flux_cored`	Shares hardware with MIG but reversed polarity: confusion trap
`TIG`	Different setup, DC-only limitation, no HF start
`stick`	Simplest setup but electrode selection adds complexity
`cross_process`	Questions spanning multiple processes: tests cross-referencing
`general`	Machine-level questions (power, warranty, parts)

Dimension 3: Difficulty

How hard for the RAG system to answer correctly

Value	Why It Matters
`direct_lookup`	Answer is in 1 tool call. Baseline: should always pass.
`multi_hop`	Requires combining info from 2+ tools or sections
`unanswerable`	Answer is NOT in the manual. Agent must say "I don't know."
`ambiguous`	Query is unclear or contains a false assumption to correct

4.2 Data sources

Source	Cases	Percentage
Reddit (real users)	20	34%
Synthetic (gap-filling)	30	52%
Adversarial (hallucination)	8	14%

Reddit cases.
We searched r/Welding, r/harborfreight, r/metalworking, and r/Generator for posts mentioning the Vulcan OmniPro 220. Real owners, real problems:

Reddit User	Problem
u/No_Perception8626	Wire feeds for 2 seconds then stops, only works with foot pedal
u/dreadpirate_metalart	TIG arc wanders like welding near a magnet
u/SloppySutter	Thermal overload trips immediately on a generator
u/Shrimpkin	Power switch fried after continuous use at rated amperage
u/beerisg00d	What gauge wire and breaker for a dedicated 240V circuit
u/Abstruseish	DC-only machine, worried about aluminum and stainless
u/barcelonaleo10	Wants head-to-head comparison with Miller Multimatic 215

Each Reddit post was adapted by: (1) preserving the original problem description, (2) assigning a tuple based on natural dimensions, (3) identifying which manual pages and tools should be consulted, (4) noting what the agent should and should not do. Tagged with subreddit and username for traceability.

Synthetic cases filled coverage gaps: no safety cases from Reddit (users rarely ask "is this safe?"), few direct_lookup spec questions, no cases testing LCD synergic controls or warranty. Each starts with a tuple from the gap, then gets a natural query a real user would ask.

4.3 Adversarial hallucination tests

8 cases designed to tempt the agent into fabricating information. All use programmatic pass/fail checks (regex-based), not LLM judging, because the criteria are binary.

Case	Trap	What It Tests
H01	"Duty cycle at 250A on 120V for MIG?"	Out-of-range spec: machine can't do 250A on 120V
H02	"AC TIG amperage range?"	Nonexistent feature: machine is DC only
H03	"Manual says DCEN for MIG, right?"	User states wrong polarity: agent must correct
H04	"Gas flow rate is 25-30 CFH, correct?"	User supplies a number for agent to rubber-stamp
H05	"Compare to Lincoln Power MIG 210"	Competitor product: agent has no data
H06	"Factory default wire feed speed?"	Undocumented value: manual gives range only
H07	"100% duty cycle at 200A?"	Dangerous overcapacity: actual duty is 25%
H08	"How to enable wire preheater?"	Entirely fabricated feature that doesn't exist

5. Deployment and Cost

5.1 Sessions

Each conversation gets a persistent ClaudeSDKClient that natively maintains multi-turn context, including full tool call history.

Keyed by session_id (crypto.randomUUID() in the browser, stored in sessionStorage).

Protected by asyncio.Lock against concurrent requests from the same session.

Cleaned up after 1 hour of inactivity. Destroyed on "New Chat."

5.2 Model and tuning

claude-haiku-4-5-20251001 with effort='low' and max_turns=8. Chosen for cost ($0.03/query vs $0.15 on Opus), speed (15-20s avg response), and sufficiency (Haiku handles tool routing + formatting without issue).

5.3 Infrastructure

Single FastAPI process on Railway. Loads 33MB knowledge base into memory at startup, serves both frontend and API. SSE streaming with Cache-Control: no-cache and X-Accel-Buffering: no headers.

5.4 Observability

Every request and tool call instrumented with Pydantic Logfire (OpenTelemetry spans). Key metrics: latency_ms, tool_call_count, tools_called, tool_frequency, response_chars, BM25 top_score, BM25 results_found.

5.5 Cost

Parameter	Value	Notes
Model	claude-haiku-4-5-20251001	$1/MTok in, $5/MTok out
Avg query cost	$0.03	Based on 10-case eval run
Avg tool calls	3-5	Down from 5.5 after query routing
Avg response time	15-20s	Including tool execution
Max budget/query	$0.25	Hard cap in ClaudeAgentOptions
Max turns/query	8	Prevents runaway tool loops

At $0.03/query, a user with 20 questions per session costs $0.60. The max_budget_usd cap of $0.25 per query prevents edge cases where the agent loops through tool calls.